

Aggregation Semantics for Prioritization in Cooperative Automated Decision Making

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Abstract

To be rewritten.

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1. Introduction

1.1. Research Questions and Contributions

2. Background and Related Work

2.1. Context-Aware and Cooperative Decision Support Systems

The increasing adoption of Artificial Intelligence (AI) in decision-support systems has highlighted the need to complement predictive performance with contextual, organizational, and domain-specific knowledge. In many real-world settings, decisions cannot be based solely on statistical predictions because institutional policies, expert knowledge, regulatory requirements, and operational constraints must also be considered. As a result, a growing body of research has focused on context-aware and cooperative decision-support systems capable of integrating heterogeneous sources of information into the decision-making process.

One important research direction concerns the incorporation of explicit knowledge representations alongside data-driven models. Early examples include cognitive-agent architectures that combine symbolic rules with machine-learning mechanisms to support context-aware decision making [1]. Similarly, ontology-based clinical decision-support systems have employed contextual ontologies and clinical pathways to select context-appropriate recommendations for individual cases [2]. More recent developments have extended these ideas through semantic models, knowledge graphs, and hybrid AI architectures that combine predictive models with structured domain knowledge to improve decision quality and interpretability [3,4].

A second research stream emphasizes cooperation between human and artificial intelligence. Rather than viewing decision support as a purely predictive task, these approaches consider decision making as a collaborative process in which machine-learning models provide evidence while human experts contribute contextual understanding, oversight, and governance. Cronholm and Göbel [5], for example, argue that effective decision support requires appropriate mechanisms for combining human and artificial intelligence rather than simply improving predictive accuracy. Likewise, recent surveys on human-centred AI highlight transparency, accountability, contextual awareness, and governance as fundamental requirements for trustworthy decision-support systems [6].

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Related ideas also appear in cooperative and distributed decision-making environments. Multi-agent systems, collective-intelligence frameworks, and cooperative AI architectures frequently rely on multiple decision sources that must be reconciled before a final recommendation is produced [7]. In these settings, contextual information often plays a coordinating role by constraining, guiding, or validating the outputs generated by predictive components. Similar concerns arise in governance-oriented decision-support systems, where contextual information may encode operational procedures, organizational priorities, safety requirements, or regulatory obligations [4,8].

Taken together, these studies clearly demonstrate the importance of contextual knowledge in decision-support systems. However, their primary focus typically lies in knowledge representation, ontology design, human–AI interaction, governance mechanisms, or system architecture. Once predictive evidence and contextual assessments have been generated, the numerical integration process that combines both sources of information is often treated as a secondary implementation detail. Consequently, relatively little attention has been devoted to understanding how alternative integration mechanisms affect the final recommendations produced by the system.

This limitation becomes particularly visible in architectures that explicitly separate predictive evidence from contextual assessment. Recent work on Cooperative Automated Decision-Making Systems (CADEMAS) [9] and its machine-learning instantiation CADEMAs-ML [10] exemplifies this design philosophy. CADEMAs organizes heterogeneous automated decision-making components and contextual knowledge into distinct layers, allowing predictive evidence and contextual assessments to be generated independently before being combined into a final recommendation. Such architectures make the integration mechanism itself an explicit design choice rather than an implicit implementation detail. As a result, the aggregation operator determines how predictive evidence interacts with contextual requirements, organizational priorities, or policy constraints, thereby shaping the semantics of the resulting decision process.

This observation suggests that aggregation should not be viewed merely as a numerical post-processing step. Instead, it can be interpreted as a governance mechanism that regulates the interaction between predictive and contextual information. Understanding the implications of different aggregation semantics therefore becomes essential for the design of cooperative and context-aware decision-support systems. The next subsection reviews the aggregation literature from this perspective.

2.2. Aggregation Operators and Decision Semantics

Aggregation operators constitute one of the central building blocks of multi-criteria decision making (MCDM), fuzzy decision support, and collective decision-making systems. Their purpose is to combine multiple sources of information into a single assessment while preserving, to a greater or lesser extent, the decision semantics associated with the original criteria. Consequently, aggregation has been studied extensively from both theoretical and applied perspectives.

Among the most influential aggregation frameworks are Ordered Weighted Averaging (OWA) operators [11,12], t-norm and t-conorm based operators, geometric aggregation functions, and non-additive integrals such as the Choquet and Sugeno integrals. Collectively, these approaches provide a rich mathematical vocabulary for representing different attitudes toward compensation, conjunction, disjunction, and interaction among criteria. From this perspective, the choice of aggregation operator is not merely a numerical convenience but a modeling decision that determines how favorable and unfavorable evidence may compensate for one another.

A substantial body of literature has explored these properties in traditional decision-making settings. Reviews by Mardani et al. [13] and Csiszár [14] highlight the widespread use of OWA-based aggregation and related fuzzy operators across diverse application domains. More recent studies have continued to investigate how aggregation semantics influence consensus formation, robustness, criterion interaction, and ranking behavior. Examples include consensus-oriented aggregation in group decision making [15,16], robust Choquet-based decision models [17,18], and applications employing geometric and OWA-type operators to capture alternative decision attitudes [19,20].

Despite this methodological richness, most aggregation studies are formulated within classical MCDM settings, where the aggregated inputs represent multiple criteria, attributes, preferences, or expert judgments associated with the same decision problem. The primary objective is typically to identify the most appropriate operator for combining these criteria or to develop new aggregation mechanisms with desirable mathematical properties. Comparatively less attention has been paid to architectures in which the aggregated inputs originate from fundamentally different decision layers.

This distinction is particularly important in cooperative automated decision-making systems. In architectures such as CADEMAs-ML, the aggregation process does not combine multiple criteria describing an alternative. Instead, it reconciles two conceptually distinct sources of evidence: a predictive assessment generated by machine-learning models and a contextual assessment encoding organizational priorities, regulatory requirements, expert knowledge, or policy constraints. Under this interpretation, the aggregation operator effectively determines the extent to which predictive evidence may compensate for weak contextual support, or conversely, whether contextual constraints can restrict otherwise favorable predictions. The operator therefore acts as an implicit governance mechanism rather than merely a criterion-combination tool.

For this reason, the present work focuses on three representative aggregation regimes exhibiting fundamentally different compensation semantics. The weighted arithmetic mean represents a fully compensatory integration strategy in which predictive and contextual information may offset one another. The weighted geometric mean represents a partially compensatory strategy that penalizes strong imbalances between both sources of evidence. Finally, the minimum operator represents a non-compensatory strategy in which the weakest component determines the final assessment. Together, these operators span a spectrum ranging from highly permissive to highly restrictive integration behavior while remaining interpretable and widely understood within the aggregation literature.

From the perspective of this study, the objective is therefore not to propose a new aggregation operator nor to extend aggregation theory. Instead, aggregation operators are treated as alternative decision semantics within a cooperative automated decision-making architecture. The key question is how different compensation regimes influence policy compliance, prioritization behavior, robustness to predictive perturbations, and ranking stability when predictive and contextual information must jointly determine the final recommendation.

2.3. Research Gap and Positioning

The reviewed literature reveals two largely independent research streams. On the one hand, context-aware and cooperative decision-support systems have developed increasingly sophisticated mechanisms for incorporating contextual knowledge into predictive processes through ontologies, symbolic rules, knowledge graphs, human oversight, and governance-oriented architectures [1,2,5,6]. These studies demonstrate the importance of contextual information but typically focus on knowledge representation, rule selection, system architecture, or human–AI interaction. Once predictive and contextual assessments

have been generated, the mechanism responsible for their numerical integration is often treated as an implementation detail.

On the other hand, aggregation theory and fuzzy MCDM have developed a rich collection of operators with diverse compensation semantics, including OWA operators, geometric means, t-norms, Choquet integrals, and Sugeno integrals [11–14]. This literature provides extensive theoretical knowledge regarding compensation, criterion interaction, robustness, consensus formation, and preference aggregation. However, aggregation operators are generally studied as mathematical constructs or as components of multicriteria decision methods, where the aggregated inputs represent criteria, preferences, or expert judgments associated with the same decision problem.

Consequently, a gap exists between context-aware decision-support frameworks and aggregation theory. The former provides mechanisms for representing contextual knowledge but rarely investigates how different aggregation semantics affect the preservation of contextual constraints after integration. The latter offers powerful aggregation mechanisms but seldom studies their implications when heterogeneous decision signals, such as predictive machine-learning outputs and contextual assessments, must jointly determine operational recommendations.

From this perspective, the gap addressed in this work is not the absence of aggregation operators, nor the lack of mathematical knowledge about their properties. Rather, it is the lack of a systematic understanding of aggregation as a governance mechanism within cooperative automated decision-making architectures. In particular, relatively little attention has been devoted to questions such as whether contextual assessments can effectively act as veto mechanisms, how different compensation regimes influence policy compliance, how sensitive prioritization outcomes are to predictive overconfidence, or how ranking stability is affected by uncertainty in contextual information.

These observations motivate the following research questions:

- *RQ1. How do different aggregation semantics affect the interaction between predictive evidence and contextual assessments in a cooperative automated decision-making architecture?*
- *RQ2. Under what conditions can aggregation operators preserve or violate contextual constraints and veto-like policies?*
- *RQ3. How do alternative aggregation regimes influence prioritization robustness under predictive overconfidence and contextual uncertainty?*
- *RQ4. To what extent do different aggregation semantics alter the stability of the resulting rankings and intervention priorities?*

Unlike traditional fuzzy MCDM studies, the objective of the present work is not to propose a new aggregation operator, derive new algebraic results, or solve a classical multicriteria decision problem. Likewise, unlike many context-aware DSS frameworks, the focus is not on developing new mechanisms for representing contextual knowledge. Instead, aggregation operators are interpreted as alternative policy semantics within the CADEMAs architecture. By keeping both the predictive subsystem and the contextual subsystem fixed, the study isolates the aggregation mechanism as an independent design variable and examines its implications for policy compliance, robustness, and prioritization behavior. This positioning distinguishes the contribution from both aggregation theory and context-aware decision-support research.

2.4. Chronological Comparison of Related Work

Table 1 summarizes representative contributions discussed in the previous subsections and positions the present work with respect to the existing literature. The comparison focuses on four dimensions that emerge from the identified research gap: (i) the explicit integration of predictive and contextual information, (ii) the treatment of contextual con-

Table 1. Chronological comparison of representative related work. “Partial” indicates that a contribution addresses the dimension only indirectly (e.g., through rules, consensus thresholds, or architectural design) without a dedicated analysis of predictive–contextual score fusion.

Reference	Year	Main focus	Predictive + contextual integration	Policy / veto analysis	Aggregation robustness	Cooperative ADM
Teng and Tan [1]	2008	Context-aware cognitive agents	✓	Partial	No	Partial
Yao and Kumar [2]	2012	Ontology-based clinical DSS	✓	Partial	No	No
Merkert et al. [21]	2015	ML in DSS survey	Partial	No	No	No
Moral et al. [15]	2017	Consensus aggregation	No	No	Partial	No
Mardani et al. [13]	2018	Fuzzy MCDM review	No	No	No	No
Csiszár [14]	2021	OWA theory	No	Partial	Partial	No
Cronholm and Göbel [5]	2022	Human-centred AI DSS	✓	Partial	No	Partial
Yang et al. [17]	2022	Robust Choquet-based MCDM	No	Partial	✓	No
Liao et al. [18]	2023	Contextual Choquet models	Partial	No	✓	No
Trillo et al. [16]	2023	Consensus-based GDM	No	Partial	No	No
Fanfani et al. [6]	2025	Human-centred AI survey	✓	Partial	No	Partial
Present work	2026	CADEMAS aggregation semantics	✓	✓	✓	✓

straints or veto-like policies, (iii) the analysis of aggregation robustness, and (iv) the study of aggregation within cooperative automated decision-making architectures.

The table illustrates that prior research has typically focused either on context-aware decision-support architectures or on aggregation methods in MCDM and fuzzy decision making. Comparatively little attention has been devoted to aggregation as a governance mechanism that simultaneously integrates predictive evidence, contextual knowledge, policy constraints, and robustness considerations within a cooperative ADM framework.

3. CADEMÁS and CADEMÁS-ML

The Cooperative Automated Decision-Making System (CADEMÁS) framework was introduced by Novoa-Hernández et al. [9] as a general architecture for context-aware decision support in which multiple heterogeneous Automated Decision-Making (ADM) units cooperate to produce a unified recommendation, score, or decision. Rather than prescribing a specific decision model, CADEMÁS provides a modular structure that separates the generation of decision evidence from its contextual evaluation and final integration.

This separation is particularly relevant for the problem studied in the present work. Many decision-support systems combine predictive evidence with contextual information, but these elements are often intertwined within the same model or decision process. In contrast, CADEMÁS explicitly distinguishes three conceptually different functions: (i)

evidence generation, (ii) contextual assessment, and (iii) decision integration. Consequently, the aggregation mechanism can be analysed independently of both the predictive subsystem and the contextual knowledge representation.

Formally, a CADEMAs instance is represented by the tuple

$$\mathcal{S} = \langle I, U, O, K, T \rangle, \quad (1)$$

where I denotes the Input Manager, U the set of ADM units, O the Decision Integration Layer, K the Contextual Modulator, and T the universe of admissible data representations.

Given an input case, the ADM units independently generate decision evidence from the available information. In parallel, the Contextual Modulator evaluates the extent to which the same case satisfies organizational priorities, expert knowledge, regulatory requirements, or other contextual constraints. The outputs of both subsystems are then combined by the Decision Integration Layer to produce the final decision score.

An important specialization of this framework is CADEMAs-ML [10], in which the ADM units correspond to machine-learning models. In this setting, multiple predictive models cooperate to produce a consolidated predictive assessment, while contextual information is represented through a fuzzy inference system inspired by Pérez-Cañedo et al. [22] and Pérez-Cañedo et al. [23]. The contextual subsystem translates organizational priorities and institutional requirements into a normalized degree of contextual satisfaction.

To focus on the role of aggregation, the internal implementation of both subsystems is abstracted in terms of two normalized scores. For each decision case i , the predictive subsystem produces a predictive score

$$R_i \in [0, 1], \quad (2)$$

which summarizes the cooperative evidence generated by the predictive ADM units. Depending on the application domain, R_i may represent a risk estimate, a confidence score, a probability of occurrence, or any other normalized predictive assessment.

Independently, the contextual subsystem produces a context score

$$Q_i \in [0, 1], \quad (3)$$

which quantifies the degree to which the case satisfies the contextual requirements encoded by the organization, institution, or application domain.

It is worth noting that the restriction of both predictive and contextual scores to the unit interval does not limit the generality of the proposed formulation. Any continuous score defined over an interval $[a, b]$ can be transformed into the normalized domain $[0, 1]$ through an affine mapping. This transformation preserves ordering relations and therefore maintains the ranking semantics required in prioritization tasks. Consequently, R_i and Q_i should be interpreted as normalized semantic scores rather than measurements expressed in a physical unit. The use of the unit interval allows the analysis of aggregation properties independently of the original scale of the predictive and contextual information. More importantly, the lower boundary $Q_i = 0$ represents a complete *contextual violation*.

The final prioritization score is then obtained through an aggregation operator

$$P_i = A(R_i, Q_i), \quad (4)$$

where A defines the integration semantics between predictive evidence and contextual assessment. Candidate cases are subsequently ranked according to P_i , and the corresponding Top- K subset is selected for intervention or resource allocation.

It is important to emphasize that the aggregation problem considered here differs from traditional fuzzy MCDM settings. In classical MCDM, aggregation operators combine multiple criteria describing different attributes of the same alternative, typically with the objective of capturing trade-offs among decision criteria. In CADEMAs-ML, however, the operator combines heterogeneous sources of decision evidence that play fundamentally different roles: a predictive assessment generated by cooperating machine-learning models and a contextual assessment derived from organizational, regulatory, or domain-specific knowledge.

Consequently, the aggregation operator does not merely determine the relative importance of criteria. Instead, it governs how predictive evidence interacts with contextual requirements and therefore acts as an explicit policy-design component within the decision architecture. The central question is not which operator produces the most desirable aggregate score in a multicriteria sense, but rather how alternative aggregation semantics affect policy compliance, contextual veto preservation, robustness to predictive overconfidence, and ranking stability under uncertainty.

Existing CADEMAs-ML applications employ a weighted linear combination of predictive and contextual scores [10]. However, the CADEMAs framework itself does not prescribe any particular aggregation operator [9]. Consequently, different aggregation semantics can be introduced without modifying either the predictive subsystem or the contextual subsystem. This modularity provides an ideal setting for studying aggregation as a policy-design mechanism and for analysing how different compensation regimes affect prioritization, policy compliance, robustness, and ranking stability. The following section introduces the aggregation operators considered in this study and develops the theoretical properties that motivate the subsequent experimental analysis.

4. Aggregation Semantics and Theoretical Analysis

The aggregation mechanism in CADEMAs-ML determines how predictive evidence and contextual knowledge are integrated into a unified prioritization decision. Therefore, the aggregation operator should not be interpreted as a mere mathematical component, but as an explicit design choice that defines the interaction between machine-learning evidence and contextual requirements.

This section provides the theoretical foundations required to answer the research questions introduced in Section 2.3, particularly those related to the role of aggregation semantics in cooperative automated decision-making. Specifically, the analysis addresses how different aggregation regimes modify the interaction between predictive and contextual information (RQ1), under which conditions contextual constraints and veto-like policies are preserved or violated (RQ2), how aggregation semantics influence robustness to predictive overconfidence and contextual uncertainty (RQ3), and how they affect the stability of the resulting prioritization rankings (RQ4).

The analysis does not aim to introduce a new aggregation operator or to extend classical aggregation theory. Instead, it studies existing aggregation paradigms as alternative policy semantics within the CADEMAs-ML architecture. The section is organized in three steps. It first formalizes the structural properties that characterize aggregation behavior, including boundedness, monotonicity, continuity, zero absorption, contextual veto preservation, and compensability. It then examines three representative aggregation regimes with different decision semantics: fully compensatory linear aggregation, sub-compensatory aggregation through the weighted geometric mean, and non-compensatory minimum aggregation. It closes by deriving conditions for contextual veto preservation, rank reversal under compensation, and sensitivity to perturbations in predictive and contextual assessments.

These results provide a formal basis for interpreting aggregation operators as policy-design components in cooperative automated decision-making systems and motivate the experimental evaluation of their practical consequences in the following sections.

4.1. Structural Properties of Aggregation Operators

Without loss of generality, all scores are assumed to be normalized within the unit interval. Any bounded continuous scoring scale $[a, b]$ can be transformed into this representation through a monotonic affine transformation, preserving the ordering relations and the aggregation semantics considered in this work.

Let

$$A : [0, 1]^2 \rightarrow [0, 1] \quad (5)$$

denote an aggregation operator that combines the predictive score R_i and the contextual score Q_i into the final prioritization score

$$P_i = A(R_i, Q_i). \quad (6)$$

Throughout this work, aggregation operators are interpreted as integration policies governing the interaction between predictive evidence and contextual knowledge within the CADEMAs-ML architecture. **Rather than focusing on specific functional forms, the following definitions introduce a set of structural properties that characterize different aggregation semantics.**

Definition 1 (Boundedness). *An aggregation operator is bounded if*

$$0 \leq A(r, q) \leq 1, \quad \forall (r, q) \in [0, 1]^2. \quad (7)$$

Boundedness guarantees that the resulting prioritization score remains within the same normalized scale as the predictive and contextual assessments. **Since A maps $[0, 1]^2$ into $[0, 1]$, this property is automatically satisfied and is stated here only for completeness.**

Definition 2 (Monotonicity). *An aggregation operator is monotonic if*

$$r_1 \leq r_2, q_1 \leq q_2 \implies A(r_1, q_1) \leq A(r_2, q_2). \quad (8)$$

Monotonicity ensures that improving either predictive evidence or contextual satisfaction cannot decrease the final prioritization score.

Definition 3 (Continuity). *An aggregation operator is continuous if small perturbations in its inputs produce arbitrarily small changes in the aggregated score.*

Continuity is particularly relevant when predictive and contextual scores are estimated from noisy data, since it prevents abrupt changes in prioritization due to infinitesimal variations of the inputs.

Definition 4 (Zero Absorption). *An aggregation operator satisfies the zero-absorption property if*

$$A(r, 0) = 0, \quad \forall r \in [0, 1]. \quad (9)$$

Under zero absorption, a null contextual assessment completely suppresses the final prioritization score regardless of the predictive evidence available.

Definition 5 (Strict positivity on the interior). *An aggregation operator is strictly positive on the interior if*

$$A(r, q) > 0, \quad \forall (r, q) \in (0, 1]^2. \quad (10)$$

Zero absorption and strict positivity play complementary roles. The first forces a null score exactly when the context vetoes an alternative. The second guarantees that no other alternative is driven to zero. Taken together, they make the value $P_i = 0$ an unambiguous marker of a contextual veto, which is what allows veto preservation to be read directly from the aggregated scores.

Definition 6 (Contextual Veto Preservation). *An aggregation operator preserves contextual vetoes if*

$$Q_i = 0 \implies P_i = 0. \quad (11)$$

Within the CADEMAs-ML setting, contextual veto preservation means that contextual requirements cannot be overridden by predictive evidence. Observe that zero absorption is a sufficient condition for contextual veto preservation. **In fact, for $Q_i = 0$, zero absorption and contextual veto preservation are equivalent.**

Definition 7 (Compensability). *In this work, an aggregation operator is compensatory on an interior region if a decrease in one input can be locally offset by an increase in the other while preserving the aggregated score. For a differentiable operator, a sufficient local characterization is that both partial derivatives are strictly positive in that region.*

Thus, compensability denotes local substitutability rather than the ability to offset an arbitrary deficit. The unit-scale bounds limit every trade-off. It determines the extent to which strong predictive evidence may compensate for weak contextual support, or vice versa. At one extreme, fully compensatory operators permit the broadest within-scale trade-offs between prediction and context. At the opposite extreme, non-compensatory operators prevent such trade-offs and therefore behave as hard contextual constraints.

These properties provide a theoretical basis for distinguishing alternative integration semantics within CADEMAs-ML. In particular, zero absorption and compensability play a central role in determining whether contextual requirements behave as soft preferences, partial constraints, or strict vetoes. The following subsection introduces the specific aggregation operators considered in this work and analyzes how they instantiate these structural properties.

4.2. Aggregation Operators Considered

The CADEMAs framework does not prescribe a specific mechanism for combining predictive and contextual assessments [9]. Consequently, different aggregation operators can be incorporated without modifying either the predictive subsystem or the contextual subsystem.

To investigate how aggregation semantics affect prioritization behavior, policy compliance, and robustness, this work considers three representative operators that span different levels of compensability between predictive evidence and contextual requirements.

1. *Linear aggregation.* The linear operator corresponds to the integration mechanism adopted in existing CADEMAs-ML implementations [10]:

$$A_L(R_i, Q_i) = \lambda R_i + (1 - \lambda)Q_i, \quad \lambda \in [0, 1]. \quad (12)$$

The parameter λ controls the relative importance assigned to predictive and contextual information. For $\lambda \in (0, 1)$, this operator is compensatory: reductions in one component can be offset by increases in the other, subject to the bounds of the unit interval. The boundary cases $\lambda = 0$ and $\lambda = 1$ are degenerate and are not considered as compensatory regimes in the subsequent analysis.

2. *Weighted geometric aggregation.* An intermediate alternative is the weighted geometric mean:

$$A_G(R_i, Q_i) = R_i^\lambda Q_i^{1-\lambda}, \quad \lambda \in (0, 1). \quad (13)$$

Unlike the linear operator, the weighted geometric mean penalizes imbalanced evaluations and therefore reduces the degree of compensation between predictive and contextual information. Nevertheless, partial compensation remains possible whenever both inputs are strictly positive. Restricting the exponent to $\lambda \in (0, 1)$ keeps both powers strictly between 0 and 1, so the indeterminate form 0^0 never arises and $A_G(r, 0) = 0$ holds for every r . The two excluded endpoints behave differently: at $\lambda = 1$ the operator reduces to $A_G(R_i, Q_i) = R_i$ and loses the veto property, and at $\lambda = 0$ it reduces to Q_i . Both are therefore excluded from the analysis, and the implementation assigns $A_G = 0$ whenever either input equals zero.

3. *Minimum aggregation.* The third regime is the minimum operator

$$A_M(R_i, Q_i) = \min(R_i, Q_i). \quad (14)$$

This operator represents a non-compensatory integration regime in which the final prioritization score cannot exceed the weakest component. Consequently, predictive evidence cannot compensate for insufficient contextual support. Note that A_M carries no weighting parameter. Its output is invariant under changes of λ , so any comparison across integration regimes leaves the minimum operator unchanged by construction. This point matters when reading the λ -dependent results of Sections 6.1 and 6.4.

Table 2 summarizes the structural properties of the three operators according to the definitions introduced in the previous subsection.

Table 2. Structural properties of the aggregation operators considered in this work.

Property	Linear	Geometric	Minimum
Boundedness	Yes	Yes	Yes
Monotonicity	Yes	Yes	Yes
Continuity	Yes	Yes	Yes
Zero absorption	No [†]	Yes	Yes
Contextual veto preservation	No [†]	Yes	Yes
Strict positivity on $(0, 1]^2$	Yes	Yes	Yes
Compensability	Full [‡]	Partial	No

[†] For the linear operator, zero absorption and contextual veto preservation hold only in the degenerate case $\lambda = 0$.

[‡] Full compensability for the linear operator is understood for $\lambda \in (0, 1)$, with compensation bounded by the $[0, 1]$ scale.

All three operators are strictly positive on the interior, so they differ only in zero absorption. This isolates the property responsible for the contrasting policy behaviour reported in Section 6: the operators agree on which alternatives deserve a positive score and disagree only on whether a contextual veto can be overridden.

These operators were selected because they represent three distinct integration semantics within CADEMAs-ML: *full within-scale* compensation, partial compensation, and strict contextual constraint. The following subsections analyze the theoretical consequences of these semantics and their implications for prioritization behavior. *Here, full within-scale compensation refers to the linear operator with $\lambda \in (0, 1)$; it does not mean unbounded compensation.*

4.3. Conditions for Contextual Veto Preservation

Within CADEMAs-ML, contextual information may represent different types of requirements, including organizational priorities, regulatory constraints, safety conditions, or expert-defined restrictions. In some applications, a contextual assessment should not behave as a soft preference but rather as a hard constraint. This situation occurs when a complete contextual violation must prevent an alternative from being prioritized, regardless of its predictive evidence.

This behavior can be characterized through the zero-absorption property introduced in Section 4.1. An aggregation operator provides contextual veto preservation if

$$A(r, 0) = 0, \quad \forall r \in [0, 1]. \quad (15)$$

Therefore, the question of whether CADEMAs-ML can enforce contextual vetoes reduces to determining whether the selected aggregation operator satisfies zero absorption.

Theorem 1 (Contextual veto preservation). *Let $A : [0, 1]^2 \rightarrow [0, 1]$ be an aggregation operator used in CADEMAs-ML. If A satisfies zero absorption, then every decision case i with a null contextual assessment,*

$$Q_i = 0, \quad (16)$$

receives a null prioritization score,

$$P_i = A(R_i, Q_i) = 0, \quad (17)$$

independently of its predictive score R_i .

Proof. The result follows directly from the definition of zero absorption. Since $Q_i = 0$, it follows that $P_i = A(R_i, Q_i) = A(R_i, 0)$. Because A satisfies zero absorption, $A(R_i, 0) = 0$, for every $R_i \in [0, 1]$. Therefore, $P_i = 0$. Hence, a complete contextual violation completely suppresses the final prioritization score regardless of the predictive evidence. $\square$

Corollary 1 (Veto preservation of the considered operators). *Among the aggregation operators considered in this work, the minimum and weighted geometric operators preserve contextual vetoes, whereas the linear operator does not for $\lambda \in (0, 1]$, with the degenerate case $\lambda = 0$ being an exception.*

Proof. For the minimum operator, $A_M(r, 0) = \min(r, 0) = 0$. Similarly, for the weighted geometric operator, $A_G(r, 0) = r^\lambda 0^{1-\lambda} = 0$, for $\lambda \in (0, 1)$. In contrast, for $\lambda \in (0, 1]$, the linear operator yields $A_L(r, 0) = \lambda r$, which is greater than zero whenever $r > 0$ and $\lambda > 0$. Therefore, predictive evidence can override a complete contextual violation under linear aggregation. For the degenerate case $\lambda = 0$, $A_L(r, 0) = 0$, so the linear operator trivially preserves contextual vetoes. $\square$

Theorem 1 constrains individual scores. Prioritization, however, is a competitive process: what matters in practice is whether a vetoed alternative can still enter the Top- K set. The following result closes that gap and identifies the extra condition required.

Corollary 2 (Exclusion of vetoed alternatives from the Top-K set). *Let A satisfy zero absorption and strict positivity on the interior (Definition 5). Consider a population of N alternatives, of which N_V are vetoed ($Q_i = 0$) and the remaining $N - N_V$ satisfy $R_i > 0$ and $Q_i > 0$. If $N - N_V \geq K$, then $\mathcal{T}_K \cap \mathcal{V} = \emptyset$ and the policy violation rate is $V = 0$.*

Proof. By zero absorption, every vetoed alternative receives $P_i = 0$. By strict positivity, each of the $N - N_V$ remaining alternatives receives $P_i > 0$. Since $N - N_V \geq K$, at least K alternatives carry a strictly positive score, so the K largest scores are all positive. No alternative with $P_i = 0$ can therefore be selected, which gives $\mathcal{T}_K \cap \mathcal{V} = \emptyset$ and $V = 0$. $\square$

Two consequences deserve attention. First, veto preservation at the score level does not by itself guarantee policy compliance at the population level: the supply condition $N - N_V \geq K$ is also needed. If vetoes were so numerous that fewer than K admissible alternatives remained, a zero-absorbing operator would be forced to fill the remaining slots with vetoed cases. Second, the corollary relies on the *true* contextual score entering the operator. When the aggregation step receives a noisy observation Q'_i instead, the hypothesis $Q_i = 0$ no longer holds for the input actually used, and the guarantee is lost. This is precisely the situation studied in Section 6.3.

The previous result establishes that contextual veto preservation is not a consequence of the predictive model or the contextual knowledge representation, but rather an emergent property of the integration semantics. Consequently, two CADEMAs-ML systems with identical predictive and contextual components may exhibit fundamentally different policy behavior depending only on the selected aggregation operator.

This observation motivates the subsequent analysis of rank reversal conditions and compensation thresholds, which characterize how much contextual improvement is required to offset predictive advantages under different aggregation regimes.

4.4. Conditions for Rank Reversal under Compensation

The previous section established that contextual veto preservation depends on the existence of a zero-absorption property. However, many aggregation operators used in decision-support systems follow a different philosophy: they allow partial compensation between the aggregated sources of evidence. In such cases, a sufficiently high value in one dimension may offset a lower value in another dimension.

This property is common in compensatory aggregation schemes, including weighted arithmetic means and several non-linear aggregation operators. Although compensation provides flexibility when predictive evidence and contextual information represent partially substitutable criteria, it also introduces the possibility that the final prioritization ranking differs from the ranking induced by predictive evidence alone.

To formalize this phenomenon, let $A : [0, 1]^2 \rightarrow [0, 1]$ be a continuous aggregation operator that is strictly increasing in both arguments. Given two alternatives i and j , assume that alternative i has stronger predictive evidence, but alternative j has better contextual alignment: $R_i > R_j$ and $Q_i < Q_j$. The final prioritization scores are then $P_i = A(R_i, Q_i)$ and $P_j = A(R_j, Q_j)$. A rank reversal occurs when $P_j > P_i$, meaning that the contextual advantage of alternative j compensates for its predictive disadvantage.

Theorem 2 (General rank reversal condition under compensation). *Let $A : [0, 1]^2 \rightarrow [0, 1]$ be a continuous aggregation operator strictly increasing in both arguments. For two alternatives i and j satisfying predictive dominance $R_i > R_j$, assume the operator provides sufficient compensatory capacity such that there exists a contextual assessment $q_{\max} \leq 1$ satisfying $A(R_j, q_{\max}) > A(R_i, Q_i)$.*

Under these conditions, a rank reversal in favour of alternative j occurs if and only if $Q_j > Q_i + \Phi_A(R_i, R_j, Q_i)$, where Φ_A is the minimum contextual improvement required to compensate for the predictive difference, implicitly defined by $A(R_j, Q_i + \Phi_A) = A(R_i, Q_i)$.

Proof. Fix $R_i > R_j$ and Q_i . Define the difference function $g(q) = A(R_j, q) - A(R_i, Q_i)$. Because A is continuous and strictly increasing in its second argument, $g(q)$ is also continuous and strictly increasing. At $q = Q_i$, $g(Q_i) = A(R_j, Q_i) - A(R_i, Q_i) < 0$ because A is strictly increasing in its first argument and $R_j < R_i$. By hypothesis, there exists $q_{\max}$ such that $g(q_{\max}) > 0$. Therefore, by the Intermediate Value Theorem, there exists a unique contextual value $q^* \in (Q_i, q_{\max})$ such that $g(q^*) = 0$, which implies $A(R_j, q^*) = A(R_i, Q_i)$. Defining the required threshold as $\Phi_A(R_i, R_j, Q_i) = q^* - Q_i$, the relation $P_j > P_i$ (or $g(Q_j) > 0$) holds if and only if $Q_j > q^*$, which is equivalent to $Q_j > Q_i + \Phi_A(R_i, R_j, Q_i)$. $\square$

Under its stated feasibility condition, the theorem shows that rank reversal need not be an artefact of a particular aggregation formula: it can arise when predictive and contextual information are allowed to compensate for one another. The specific threshold Φ_A depends on the operator and on the two alternatives being compared. The theorem does not imply that every compensatory operator, or every pair of alternatives, admits a reversal within the unit interval.

This distinction is particularly relevant in CADEMAs-ML because the two aggregated components do not represent equivalent evaluation criteria. The predictive score R_i summarizes evidence generated by machine learning models, whereas the contextual score Q_i represents compliance with organizational, regulatory, or domain-specific requirements. Therefore, the choice of aggregation operator implicitly defines the degree to which contextual requirements can modify or override data-driven recommendations.

4.5. Sensitivity to Predictive and Contextual Perturbations

The previous sections analyzed the structural properties of aggregation operators in terms of compensation and contextual veto preservation. However, in practical decision-support systems, both predictive scores and contextual assessments are subject to uncertainty. Predictive models may produce unstable estimates due to limited data, model uncertainty, or distribution changes, whereas contextual assessments may vary because of incomplete knowledge, expert disagreement, or evolving organizational requirements.

Therefore, an additional relevant property of an aggregation mechanism is its sensitivity to perturbations in its input dimensions. In the context of CADEMAs-ML, this property determines how strongly changes in predictive evidence or contextual requirements affect the final prioritization score.

Let $A : [0, 1]^2 \rightarrow [0, 1]$ be an aggregation operator. Consider a perturbation ϵ applied to the contextual assessment while the predictive score remains fixed. The resulting variation in the prioritization score is $\Delta_Q(\epsilon) = |A(R, Q + \epsilon) - A(R, Q)|$. Similarly, a perturbation in the predictive assessment produces $\Delta_R(\epsilon) = |A(R + \epsilon, Q) - A(R, Q)|$. These quantities measure the local influence of each information source on the final decision score. Operators that strongly amplify small perturbations may lead to unstable rankings, whereas operators with bounded sensitivity provide greater robustness against uncertainty in predictive or contextual information.

Theorem 3 (Lipschitz sensitivity of aggregation operators). *Let $A : [0, 1]^2 \rightarrow [0, 1]$ be an aggregation operator continuously differentiable on an open neighbourhood containing the line*

segment between (R, Q) and $(R + \epsilon_R, Q + \epsilon_Q)$. Assume that both endpoints, and hence the whole segment by convexity, lie in $[0, 1]^2$. If there exist finite constants L_R and L_Q such that

$$|\partial A / \partial R| \leq L_R \quad \text{and} \quad |\partial A / \partial Q| \leq L_Q,$$

then, for any perturbations ϵ_R and ϵ_Q ,

$$|A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q)| \leq L_R |\epsilon_R| + L_Q |\epsilon_Q|.$$

Proof. By the Mean Value Theorem, there exists a point (ξ_R, ξ_Q) between (R, Q) and $(R + \epsilon_R, Q + \epsilon_Q)$ such that

$$A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q) = (\partial A / \partial R)(\xi_R, \xi_Q) \epsilon_R + (\partial A / \partial Q)(\xi_R, \xi_Q) \epsilon_Q.$$

Taking absolute values and applying the derivative bounds yields

$$|A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q)| \leq L_R |\epsilon_R| + L_Q |\epsilon_Q|.$$

□

The theorem establishes a local robustness bound. The constants L_R and L_Q quantify the maximum amplification of perturbations on the particular region considered; they do not by themselves establish a global ordering of operators or the stability of a population-level Top- K decision. Those questions are examined empirically in Section 6.

For the linear aggregation operator, $A_L(R, Q) = \lambda R + (1 - \lambda)Q$, the sensitivity coefficients are constant: $L_R = \lambda$ and $L_Q = 1 - \lambda$. Therefore, the influence of predictive and contextual uncertainty is uniform throughout the decision space. This property explains the fully compensatory behaviour of the linear operator: a perturbation in one dimension has exactly the same marginal effect regardless of the current values of prediction and context.

For the weighted geometric operator, $A_G(R, Q) = R^\lambda Q^{1-\lambda}$, the local sensitivities are $\partial A_G / \partial R = \lambda R^{\lambda-1} Q^{1-\lambda}$ and $\partial A_G / \partial Q = (1 - \lambda) R^\lambda Q^{-\lambda}$. Unlike the linear operator, these sensitivities depend on the current decision state. In particular, $\partial A_G / \partial Q$ becomes unbounded as $Q \rightarrow 0^+$ when $R > 0$. Thus, finite Lipschitz constants for A_G are available only on regions bounded away from the coordinate axes. Near a contextual veto, a small positive observational error can therefore have a large local effect on the aggregated score. This fact is consistent with, but does not by itself determine, the false-negative-veto mechanism examined in Section 6.3.

For the minimum operator, $A_M(R, Q) = \min(R, Q)$, the final score is completely determined by the limiting component. Consequently, the operator exhibits zero compensation: improving the non-limiting input has no effect until it becomes the active constraint. This property provides maximal preservation of contextual restrictions but may reduce responsiveness when contextual assessments are uncertain. Because A_M is not differentiable on $R = Q$, it is not covered by Theorem 3 there; the following corollary captures one of its piecewise-stable regimes.

Corollary 3 (Minimum operator stability under inactive-component perturbations). Consider a decision case evaluated under the minimum operator $A_M(R, Q) = \min(R, Q)$. If the predictive score acts as the strict limiting component ($R < Q$), the operator completely absorbs any

contextual perturbation ε as long as the ordinal relationship is maintained (i.e., $R \leq Q + \varepsilon$). Under these conditions, the sensitivity is zero:

$$|A_M(R, Q + \varepsilon) - A_M(R, Q)| = |R - R| = 0.$$

Proof. Let the initial state of the decision case be defined by R and Q . By hypothesis, the predictive score is the strict limiting component, meaning $R < Q$. By the definition of the minimum operator, the initial aggregated score is:

$$A_M(R, Q) = \min(R, Q) = R.$$

Now, consider a perturbed contextual assessment $Q' = Q + \varepsilon$. By hypothesis, the perturbation ε (whether positive or negative) is bounded so that the ordinal relationship is maintained, that is, $R \leq Q + \varepsilon$. Evaluating the minimum operator under this perturbed state yields:

$$A_M(R, Q + \varepsilon) = \min(R, Q + \varepsilon) = R.$$

Calculating the absolute difference between the perturbed and initial states gives the sensitivity:

$$|A_M(R, Q + \varepsilon) - A_M(R, Q)| = |R - R| = 0.$$

Thus, the perturbation is completely absorbed, and the final prioritization score remains unaffected. $\square$

This result illustrates the asymmetric robustness of non-compensatory operators. Perturbations of the non-limiting component are absorbed entirely, whereas perturbations of the active constraint pass through to the aggregated score. The transmission is never amplified: since $|\min(R, Q + \varepsilon) - \min(R, Q)| \leq |\varepsilon|$ for all R, Q, ε , the minimum operator is globally 1-Lipschitz in each argument, and its sensitivity therefore alternates between 0 and at most $|\varepsilon|$. The contrast with A_G is worth noting: the geometric operator is smooth but its contextual sensitivity grows without bound as $Q \rightarrow 0^+$, whereas A_M is non-smooth yet uniformly bounded. Smoothness and robustness are thus not equivalent here. These are score-level bounds, however, and they do not by themselves determine how stable the induced ranking is; that question is addressed empirically in Section 6.4.

These analytical differences motivate the robustness experiments conducted in the following sections. Rather than evaluating aggregation operators only according to final ranking performance, the experiments examine how their semantic properties influence stability under perturbations of predictive evidence and contextual assessments.

5. Experimental Methodology

The theoretical analysis established that aggregation operators induce different decision semantics within CADEMAs-ML, affecting compensation, contextual constraint preservation, and sensitivity to uncertainty. However, the practical consequences of these properties emerge at the system level, where multiple decision cases compete for prioritization and intervention.

This section describes the experimental methodology designed to analyze these effects. The experiments do not aim to verify the theoretical results, but rather to investigate how different aggregation semantics translate into observable decision-making behaviors under realistic population-level scenarios. The evaluation covers four outcomes: policy compliance, robustness against predictive overconfidence, sensitivity to contextual uncertainty, and ranking stability.

The experimental design follows a Monte Carlo simulation framework in which synthetic decision populations are generated under controlled conditions. This approach allows the independent manipulation of predictive evidence, contextual assessments, uncertainty levels, and aggregation regimes while keeping the underlying decision architecture unchanged. Consequently, differences observed across scenarios can be attributed to the integration semantics rather than to variations in the predictive or contextual subsystems.

5.1. Objectives and Experimental Logic

The experimental evaluation is designed to answer how alternative aggregation semantics affect the operational behavior of CADEMAs-ML when applied to collections of competing decision cases. While the theoretical analysis characterizes individual properties of aggregation operators, the experimental analysis examines their emergent consequences when scores are used to prioritize a population of alternatives.

Each of the four outcomes listed above is probed by a dedicated experiment. Policy compliance is measured by checking whether alternatives that violate contextual requirements can still reach the highest-priority positions. Predictive overconfidence is modelled by making machine-learning evidence systematically optimistic for cases with poor contextual alignment. Contextual uncertainty is introduced through controlled perturbations of the contextual assessments. Ranking stability is captured by rank-correlation and Top-K overlap measures.

For each simulated scenario, a population of decision cases is generated, where each case i is characterized by a predictive score $R_i \in [0, 1]$ and a contextual score $Q_i \in [0, 1]$. A given aggregation operator A produces the final prioritization score

$$P_i = A(R_i, Q_i),$$

which determines the ranking of alternatives and the selection of the Top-K candidates. The same population and perturbation conditions are evaluated under all aggregation regimes, allowing a controlled comparison of their decision semantics. The aggregation parameter λ is varied according to the specific operator definition, allowing the influence of predictive and contextual information to be systematically analyzed while preserving the normalized decision scale.

5.2. Synthetic Population Generation

The objective of the synthetic experiments is not to reproduce a particular application domain, but to create controlled decision environments in which the effects of different aggregation semantics can be isolated. Therefore, synthetic populations are generated by directly sampling the two components that define each CADEMAs-ML decision case: the predictive assessment and the contextual assessment.

Each simulated population consists of N independent decision cases. For each case i , a predictive score $R_i \in [0, 1]$ and a contextual assessment $Q_i \in [0, 1]$ are generated. These values represent the outputs of the predictive and contextual subsystems, respectively, while the internal mechanisms used to obtain them are abstracted away. This design allows the aggregation stage to be evaluated independently from any specific machine-learning model or knowledge representation approach.

To generate realistic score distributions, predictive and contextual values are sampled from Beta distributions:

$$R_i \sim \text{Beta}(\alpha_R, \beta_R), \quad Q_i \sim \text{Beta}(\alpha_Q, \beta_Q), \quad (18)$$

which provide flexible control over concentration, dispersion, and asymmetry within the unit interval. The Beta distribution is particularly suitable for normalized confidence, risk, and satisfaction scores because it naturally represents bounded continuous assessments.

Different experimental scenarios are obtained by modifying the parameters of these distributions and by introducing controlled perturbations or dependencies between predictive and contextual assessments when required. In particular, three main scenarios are considered:

1. *Baseline decision environment.* Predictive and contextual assessments are generated independently, representing situations where machine-learning evidence and contextual knowledge provide complementary but unrelated information.
2. *Predictive overconfidence scenario.* Predictive overconfidence is modeled as a systematic upward bias in the predictive assessment. Specifically, the original predictive score is modified as

$$R'_i = \begin{cases} \text{clip}(R_i + \delta, 0, 1), & Q_i \leq q_{\text{weak}}, \\ R_i, & Q_i > q_{\text{weak}}, \end{cases} \quad (19)$$

where $\delta \geq 0$ is an experimental parameter controlling the magnitude of predictive overconfidence, $q_{\text{weak}} = 0.25$ defines the weak-context threshold, and $\text{clip}(\cdot)$ ensures that perturbed values remain within the normalized domain $[0, 1]$. The perturbation therefore affects all contextually weak alternatives, including vetoes ($Q_i = 0$) and weak non-veto cases ($0 < Q_i \leq q_{\text{weak}}$), while leaving alternatives with $Q_i > q_{\text{weak}}$ unchanged. This scenario represents situations in which machine-learning models assign excessively high predictive scores on contextually weak cases.

3. *Contextual uncertainty scenario.* Contextual uncertainty is simulated by introducing stochastic perturbations into the contextual assessments:

$$Q'_i = \text{clip}(Q_i + \epsilon_i, 0, 1), \quad (20)$$

where

$$\epsilon_i \sim \mathcal{N}(0, \sigma_Q^2) \quad (21)$$

represents contextual uncertainty and σ_Q controls its intensity. The clipping operation guarantees that perturbed contextual assessments remain within the valid normalized range. This formulation models uncertainty arising from incomplete contextual knowledge, expert disagreement, or variations in organizational requirements.

4. *Predictive-score uncertainty scenario.* The last scenario perturbs the predictive side instead, using noise with zero mean:

$$R'_i = \text{clip}(R_i + \epsilon_i, 0, 1), \quad \epsilon_i \sim \mathcal{N}(0, \sigma_R^2), \quad (22)$$

where σ_R controls the noise level. Two features separate this scenario from predictive overconfidence. The perturbation is applied to every alternative rather than only to contextually weak ones, and it is random with zero mean rather than a systematic upward shift. It therefore represents estimation error in the predictive subsystem, not a bias in the direction of optimism. This scenario supports the stability analysis of Section 6.4.

For each scenario, identical synthetic populations are evaluated using all aggregation operators. This paired experimental design ensures that observed differences in prioritiza-

tion, policy compliance, and ranking stability are attributable to the aggregation semantics rather than to variations in the generated decision cases.

All simulations are repeated using multiple random seeds to account for sampling variability. The number of generated cases, distribution parameters, perturbation levels, and random seeds are reported in the following subsection to ensure experimental reproducibility.

5.3. Monte Carlo Protocol

For each experimental scenario described in the previous subsection, the evaluation is performed using a Monte Carlo simulation protocol. Each simulation consists of generating a synthetic population, computing the prioritization scores under each aggregation operator, and evaluating the corresponding performance metrics.

To ensure a fair comparison, every aggregation operator is evaluated on exactly the same synthetic population and under the same perturbation conditions. Consequently, any observed differences are exclusively attributable to the aggregation semantics rather than to stochastic variation in the generated data.

Each experimental configuration is replicated 1000 times, using the registered pseudo-random initializations 42–1041. The same generated population is used for all operators within a replication, so operator comparisons are paired.

Let $m_1, \dots, m_B$ denote the value of a metric obtained in each of the $B = 1000$ replications. Reported point estimates are the sample mean $\bar{m} = B^{-1} \sum_b m_b$. The accompanying interval is the pair of empirical percentiles $(m_{(2.5\%)}, m_{(97.5\%)})$ of the same sample. Such an interval describes the dispersion of the metric across replications, that is, the range within which the outcome of a single simulated population falls 95% of the time. It is therefore wider than a confidence interval for the mean, which with $B = 1000$ replications would be narrower by roughly a factor of $\sqrt{B}$. Percentile intervals are reported because the quantity of interest is the variability of the decision outcome itself, not the precision with which its average has been estimated. All figures and tables use this convention, and the term *percentile interval* is used throughout to avoid ambiguity.

Table 3 summarizes the experimental configuration adopted throughout the simulation study. These parameters remain fixed unless otherwise stated for a particular experiment.

Table 3. Monte Carlo simulation parameters.

Parameter	Value(s)
Population size (N)	1000
Monte Carlo replications	1000
Random-number initializations	1000 registered seeds (42–1041)
Aggregation weight (λ)	Experiment-specific grids; see Sections 6.1, 6.3, and 6.4
Predictive overconfidence (δ)	$\{0.00, 0.05, \dots, 0.30\}$
Weak-context threshold (q_{weak})	0.25
Contextual uncertainty (σ_Q)	$\{0.00, 0.05, 0.10, 0.15, 0.20\}$ (Experiment 03)
Predictive-score uncertainty (σ_R)	$\{0.00, 0.02, 0.05, 0.10, 0.20\}$ (Experiment 04)
Predictive distribution	Beta(2, 2) (veto group: Beta(8, 2))
Contextual distribution	Beta(2, 2) (veto group: $Q_i = 0$)
Veto fraction	0.20
Top- K selection	$K = 100$
Confidence level	95% percentile interval
Number of aggregation operators	3

5.4. Evaluation Metrics

The experimental evaluation aims to quantify the practical consequences of different aggregation semantics from four complementary perspectives: preservation of contextual policies, predictive performance, ranking stability, and robustness under uncertainty. Together, these metrics provide empirical evidence for answering the research questions formulated in Section 2.3.

5.4.1. Policy compliance

The primary objective of CADEMAs-ML is not only to prioritize high-scoring cases but also to preserve contextual requirements imposed by organizational policies or domain knowledge. The first family of metrics therefore measures how well each aggregation operator preserves contextual vetoes.

Let $\mathcal{T}_K$ denote the selected Top- K set and let $\mathcal{V} = \{i : Q_i = 0\}$ be the set of contextually invalid alternatives. The policy violation rate is defined as

$$V = \frac{|\mathcal{T}_K \cap \mathcal{V}|}{K}, \quad (23)$$

which represents the proportion of selected alternatives violating a contextual veto. The metric counts only alternatives with $Q_i = 0$ and therefore does not reflect occupancy by weak non-veto cases ($0 < Q_i \leq q_{\text{weak}}$). Lower values indicate better policy compliance, while $V = 0$ corresponds to perfect preservation of contextual constraints. A complementary quantity is also reported,

$$\text{VPR} = 1 - V, \quad (24)$$

referred to as the *Veto Preservation Rate*. Higher values indicate better preservation of contextual requirements, with $\text{VPR} = 1$ representing perfect policy compliance.

Since the Top- K size is fixed throughout the experiments ($K > 0$), both metrics are always well defined.

5.4.2. Predictive utility

Aggregation modifies the original predictive ranking by incorporating contextual information. To quantify how much predictive evidence survives aggregation, the average predictive score of the selected alternatives is computed as

$$\bar{R} = \frac{1}{K} \sum_{i \in \mathcal{T}_K} R_i. \quad (25)$$

Higher values indicate that the selected candidates remain strongly supported by the predictive subsystem. The average deliberately runs over the whole Top- K set and not only over its policy-compliant subset. Averaging over $\mathcal{T}_K \setminus \mathcal{V}$ would leave the metric undefined whenever every selected alternative is vetoed, since the denominator would vanish. With the present definition the denominator is the fixed constant $K > 0$, so $\bar{R}$ is well defined for every operator, every parameter setting, and every replication. The choice also keeps the interpretation simple: $\bar{R}$ and V describe the same selected set, one in terms of predictive quality and the other in terms of policy compliance, and can therefore be read side by side.

5.4.3. Ranking stability

To quantify the stability of an ordering under a specified perturbation, a reference score vector $P^{(1)}$ is compared with a perturbed score vector $P^{(2)}$. Kendall's τ_b measures agreement in their aggregated-score order; tied scores receive average ranks, as in the standard tie-corrected definition. The Jaccard similarity coefficient evaluates overlap between the corresponding selected Top- K sets:

$$J = \frac{|\mathcal{T}_K^{(1)} \cap \mathcal{T}_K^{(2)}|}{|\mathcal{T}_K^{(1)} \cup \mathcal{T}_K^{(2)}|}. \quad (26)$$

Kendall's τ_b therefore captures global agreement in the aggregated-score order, whereas the Jaccard index captures stability of the selected intervention set. For Top- K selection, ties in P are resolved using the predictive score available in the relevant condition as a secondary criterion and the case identifier as a final deterministic criterion. This rule guarantees reproducible selections.

5.4.4. Robustness under perturbations

Robustness is evaluated by comparing the previous metrics before and after introducing predictive overconfidence or contextual uncertainty. For each perturbation level, changes in policy violation, veto preservation, predictive utility, and ranking similarity are computed relative to the corresponding baseline scenario.

Aggregation operators exhibiting smaller degradations under increasing perturbation levels are regarded as more robust with respect to predictive bias and contextual uncertainty.

6. Simulation Results

This section presents the results obtained from the Monte Carlo experiments described in the previous section. The objective is not to validate the theoretical properties established in Section 4, but to investigate their practical consequences at the population level. In particular, the experiments examine how different aggregation semantics influence policy compliance, predictive utility, ranking stability, and robustness under predictive overconfidence and contextual uncertainty.

Unless otherwise stated, all aggregation operators are evaluated under identical experimental conditions, allowing the observed differences to be attributed exclusively to their integration semantics.

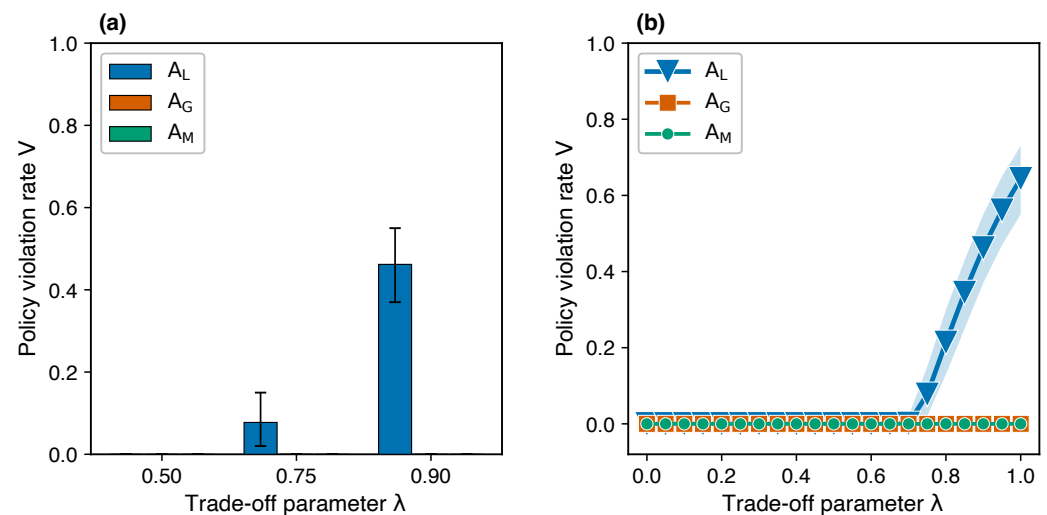


Figure 1. Policy violation rate V under the adversarial contextual-veto scenario ($N = 1000$, $K = 100$, 1000 Monte Carlo replications). The veto group consists of alternatives with $Q_i = 0$ and intentionally high predictive scores in order to create a conflict between predictive evidence and contextual constraints. Points denote Monte Carlo means; error bars and shaded bands represent 95% percentile intervals. (a) Representative integration regimes $\lambda \in \{0.50, 0.75, 0.90\}$. (b) Dense sweep over $\lambda \in [0, 1]$.

6.1. Policy Compliance under Contextual Vetoes

The first experiment evaluates the population-level consequences of the contextual veto property established in Section 4.3. Rather than comparing the overall quality of different aggregation operators, the objective is to examine whether the integration semantics preserve organizational constraints when predictive evidence conflicts with contextual requirements.

To create a controlled stress scenario, a fraction of the simulated decision cases is assigned a null contextual assessment ($Q_i = 0$), representing alternatives that should be excluded by policy. These vetoed cases are intentionally generated with high predictive scores, thereby creating an adversarial situation in which predictive evidence strongly conflicts with contextual constraints. Such a design allows the aggregation mechanism to be evaluated under the most challenging conditions for policy preservation.

Figure 1 reports the policy violation rate V under this adversarial design. Panel (a) compares the three aggregation operators at $\lambda \in \{0.50, 0.75, 0.90\}$, while panel (b) shows the full sweep over $\lambda \in [0, 1]$. Table 4 reports the corresponding Monte Carlo means, standard deviations, and 95% percentile intervals.

At $\lambda = 0.50$, no policy violations are observed for any of the aggregation operators. Under balanced integration, predictive evidence alone is insufficient to promote contextually excluded alternatives into the Top- K selection. As the predictive weight increases, however, the behavior of the linear operator changes substantially. At $\lambda = 0.75$, the average policy violation rate reaches $V = 0.077$ (95% percentile interval [0.020, 0.150]), while at $\lambda = 0.90$ it increases to $V = 0.462$ (95% percentile interval [0.370, 0.550]), indicating that nearly half of the selected interventions correspond to alternatives that violate the contextual veto. These results demonstrate that policy violations are not an intrinsic property of linear aggregation, but rather emerge progressively as predictive evidence becomes increasingly dominant.

In contrast, both the weighted geometric and minimum operators maintain $V = 0$ throughout all representative integration regimes. This outcome is not an empirical discovery but an instance of Corollary 2: the design contains 800 non-vetoed alternatives and

$K = 100$, so the supply condition $N - N_V \geq K$ holds and $V = 0$ follows deterministically. Reporting it serves as a validation check on the implementation rather than as evidence.

The informative part of the experiment lies elsewhere. Corollary 2 says nothing about the linear operator, whose violation rate is not fixed by algebra but depends on how predictive and contextual scores are distributed across the population. The simulation quantifies that dependence: it locates the value of λ at which violations first appear, and it measures how steeply they grow beyond it. Neither quantity can be read off the operator definitions, and both are what a practitioner setting λ would need to know.

Table 4. Monte Carlo estimates of the policy violation rate V under the adversarial contextual-veto scenario for representative integration regimes. Results correspond to 1000 replications with $N = 1000$ decision cases and Top- $K = 100$.

λ	Operator	V mean	V std	V $P_{2.5}$	V $P_{97.5}$
0.50	A_L (Linear)	0.000	0.000	0.000	0.000
	A_G (Geometric)	0.000	0.000	0.000	0.000
	A_M (Minimum)	0.000	0.000	0.000	0.000
0.75	A_L (Linear)	0.077	0.036	0.020	0.150
	A_G (Geometric)	0.000	0.000	0.000	0.000
	A_M (Minimum)	0.000	0.000	0.000	0.000
0.90	A_L (Linear)	0.462	0.047	0.370	0.550
	A_G (Geometric)	0.000	0.000	0.000	0.000
	A_M (Minimum)	0.000	0.000	0.000	0.000

The dense parameter sweep in panel (b) of Figure 1 further illustrates this behavior. For the simulated population considered in this experiment, the average policy violation rate of the linear operator remains zero over most of the explored parameter range and first becomes positive, for the simulated population considered here, at approximately $\lambda = 0.70$, after which it increases monotonically as more weight is assigned to predictive evidence. Conversely, the weighted geometric and minimum operators preserve complete contextual compliance over the entire range of λ , demonstrating that their veto-preserving behavior is independent of the relative weighting assigned to prediction and context.

These results illustrate Theorem 1 and Corollary 1 at the population level. Under the abundance-of-positive-scores condition built into this design, the zero-absorbing operators exclude vetoed alternatives from the Top- K set, whereas the linear operator increasingly admits them as predictive evidence receives greater weight. Thus, otherwise identical CADEMAs-ML systems can exhibit different policy behaviour solely because of the operator used to integrate the two information sources.

6.2. Effects of Predictive Overconfidence

The previous experiment showed that, under the adversarial veto design, the linear operator may select contextually vetoed alternatives when predictive evidence is given sufficient weight. At $\lambda = 0.75$, for example, the policy violation rate is already $V = 0.077$ in the absence of overconfidence. The present experiment asks how that behaviour changes when the predictive scores of contextually weak alternatives are systematically inflated. The question matters because such overconfidence reaches beyond alternatives that are explicitly vetoed: it also touches cases that the context supports only weakly without ruling them out.

The perturbation follows the overconfidence model of Section 5.2 with $q_{\text{weak}} = 0.25$, which separates two types of contextually weak alternatives: cases with $Q_i = 0$ are contextual vetoes, and cases with $0 < Q_i \leq q_{\text{weak}}$ are weak non-vetoes. Only the latter

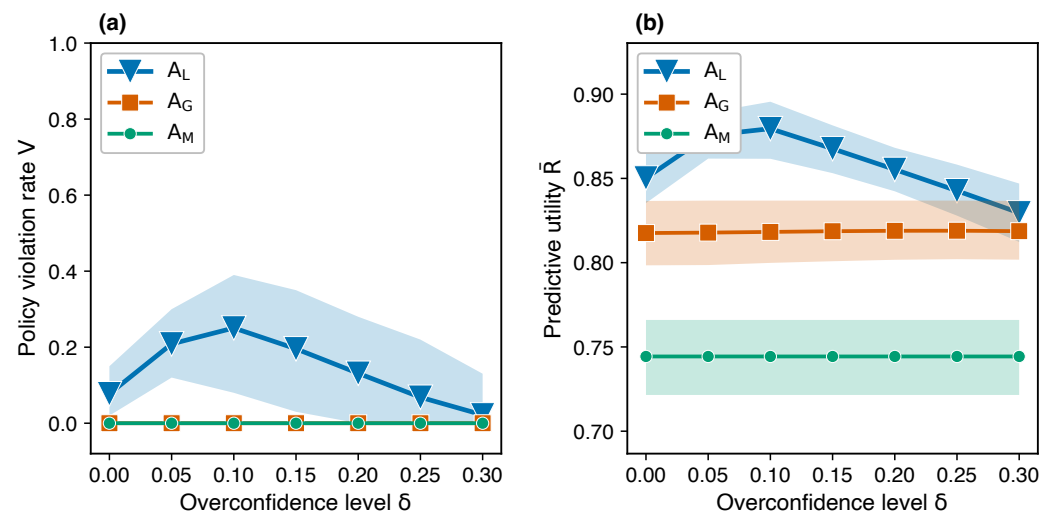


Figure 2. Policy violation rate V and mean original predictive score $\bar{R}$ (mean original R of the selected Top- K) versus overconfidence level δ at $\lambda = 0.75$ ($N = 1000$, $K = 100$, $q_{\text{weak}} = 0.25$, 1000 Monte Carlo replications). Points are means and shaded bands denote 95% percentile intervals. (a) $V(\delta)$ for A_L , A_G , and A_M . (b) $\bar{R}(\delta)$. Overconfidence is applied whenever $Q_i \leq q_{\text{weak}}$.

group receives the optimistic shift. For each $\delta \in \{0.00, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$, prioritization is performed using $P_i = A(R'_i, Q_i)$ on the same synthetic populations as in Section 6.1, and results are reported at $\lambda = 0.75$ unless stated otherwise. One detail matters for reading the results: the policy violation rate V counts only selected alternatives with $Q_i = 0$, so weak non-vetoes never contribute to V even though they remain contextually weak.

Figure 2 and Table 5 summarize the primary Monte Carlo results. For the linear operator, the effect of overconfidence is clearly non-monotonic. The violation rate increases from $V = 0.077$ at $\delta = 0.00$ to a maximum of $V = 0.251$ at $\delta = 0.10$, but then falls to $V = 0.022$ at $\delta = 0.30$. Thus, increasing predictive overconfidence does not simply produce progressively more policy violations. The mean original predictive score of the selected Top- K set, $\bar{R}$, follows a similar pattern: it increases from 0.850 at $\delta = 0.00$ to 0.880 at $\delta = 0.10$, before declining to 0.830 at $\delta = 0.30$. By contrast, the weighted geometric and minimum operators maintain $V = 0$ throughout the explored δ grid, while their mean original predictive scores remain essentially unchanged.

Table 5. Policy violation rate V and mean original predictive score $\bar{R}$ (mean R of the Top- K set) under predictive overconfidence at $\lambda = 0.75$. Overconfidence $R' = \text{clip}(R + \delta, 0, 1)$ is applied when $Q_i \leq 0.25$. Values are Monte Carlo means and 95% percentile confidence intervals over 1000 replications ($N = 1000$, $K = 100$).

δ	Operator	V mean	V std	V CI low	V CI high	$\bar{R}$ mean	$\bar{R}$ CI low	$\bar{R}$ CI high
0.00	A_L (Linear)	0.077	0.036	0.020	0.150	0.850	0.835	0.866
	A_G (Geometric)	0.000	0.000	0.000	0.000	0.818	0.798	0.837
	A_M (Minimum)	0.000	0.000	0.000	0.000	0.744	0.721	0.766
0.10	A_L (Linear)	0.251	0.078	0.080	0.390	0.880	0.862	0.896
	A_G (Geometric)	0.000	0.000	0.000	0.000	0.818	0.800	0.837
	A_M (Minimum)	0.000	0.000	0.000	0.000	0.744	0.721	0.766
0.20	A_L (Linear)	0.131	0.076	0.000	0.280	0.855	0.842	0.868
	A_G (Geometric)	0.000	0.000	0.000	0.000	0.819	0.802	0.837
	A_M (Minimum)	0.000	0.000	0.000	0.000	0.744	0.721	0.766
0.30	A_L (Linear)	0.022	0.039	0.000	0.130	0.830	0.812	0.847
	A_G (Geometric)	0.000	0.000	0.000	0.000	0.819	0.802	0.837
	A_M (Minimum)	0.000	0.000	0.000	0.000	0.744	0.721	0.766

The decrease in V at larger values of δ , however, should not be interpreted as a recovery of contextual compliance. The reason is that the composition of the Top- K set changes. As overconfidence increases beyond the peak, selected vetoes are progressively replaced by weak non-vetoes. From $\delta = 0.10$ to $\delta = 0.30$, the mean number of vetoes in Top- K falls from 25.1 to just 2.2, whereas the number of weak non-vetoes rises from 8.0 to 34.4. At the same time, total weak-context occupancy, E_{weak} , increases from 0.331 to 0.366. Thus, the decline in V occurs even though the overall presence of contextually weak alternatives in Top- K becomes slightly larger. In other words, the system is not becoming more contextually aligned; rather, the type of weak-context alternatives occupying the selected set is changing. Weak non-vetoes remain contextually weak ($Q_i \leq q_{\text{weak}}$); they are simply not counted by V because their contextual score is strictly positive.

This replacement can be understood directly from the scoring rule of the linear operator. For $A_L(R', Q) = 0.75R' + 0.25Q$, a saturated veto with $R' = 1$ and $Q = 0$ receives

$$P_v = 0.75,$$

whereas a saturated weak non-veto with $R' = 1$ and $Q > 0$ receives

$$P_w = 0.75 + 0.25Q > 0.75.$$

Once both alternatives reach the clipping boundary, therefore, every saturated weak non-veto strictly outranks every saturated veto, regardless of how close their predictive scores were before perturbation. This effect is reinforced by the different predictive score distributions used to generate the two groups: vetoes start with higher predictive scores, $R \sim \text{Beta}(8, 2)$ with mean approximately 0.80, whereas weak non-vetoes are generated from $R \sim \text{Beta}(2, 2)$ with mean approximately 0.50. Vetoes thus reach the clipping boundary earlier. Once saturated, further increases in δ can no longer increase their predictive contribution to P_v , while weak non-vetoes retain their positive contextual contribution. Among veto–weak-non-veto pairs for which both alternatives have $R' = 1$, the fraction satisfying $P_w > P_v$ is 1.000 for every tested $\delta \geq 0.05$, providing direct empirical confirmation of this ranking mechanism.

The role of clipping is also essential. On the same populations and random seeds, an unclipped counterfactual in which $R'_i = R_i + \delta$ for $Q_i \leq q_{\text{weak}}$ produces a monotonically increasing violation rate, reaching $V = 0.784$ at $\delta = 0.30$. The downturn observed in the formal experiment therefore arises from the interaction between predictive overconfidence and the bounded $[0, 1]$ predictive scale, rather than from overconfidence alone. The full clipped–unclipped comparison is reported in Appendix A.1.

The same qualitative pattern also appears when the relative weight of predictive evidence is changed, although its magnitude depends on λ . At $\lambda = 0.50$, no violations are observed over the explored δ grid. At $\lambda = 0.90$, $V(\delta)$ is again non-monotonic, reaching a maximum of $V = 0.809$ at $\delta = 0.15$. Here the same replacement mechanism operates, but from a substantially higher baseline violation rate. Overall, these results show that the effect of predictive overconfidence is not adequately described by a simple monotonic increase in policy violations: under the linear operator, clipping can first increase the selection of vetoed alternatives and then, after saturation, shift the Top-K composition toward weak non-vetoes.

6.3. Robustness to Contextual Uncertainty

The previous experiments assumed that contextual assessments Q_i are known exactly. Experiment 03 relaxes this assumption by introducing uncertainty in the contextual subsystem and studying how imperfect observations affect policy compliance. Its $\sigma_Q = 0$ sweep serves as a replication check of Section 6.1, rather than as a new compliance result: the linear operator exhibits violations for sufficiently large predictive weights, whereas the weighted geometric and minimum operators remain zero-absorbing with respect to the observed contextual input.

Contextual uncertainty is modeled as a stochastic observation error rather than as an adversarial perturbation. The observed contextual score is defined as

$$Q'_i = \text{clip}(Q_i + \epsilon_i, 0, 1),$$

where $\epsilon_i \sim \mathcal{N}(0, \sigma_Q^2)$ and σ_Q controls the uncertainty level. Aggregation is performed using the observed contextual score Q'_i , while the policy violation rate V is always computed using the true contextual state Q_i . Therefore, violations measure genuine compliance failures caused by imperfect contextual information rather than errors in the evaluation metric. This setting differs from the predictive overconfidence experiment in Section 6.2, where the uncertainty affected predictive evidence rather than contextual assessment. The additive error is zero-mean before clipping; because the score scale is bounded, Q'_i is not conditionally unbiased at the boundaries.

Figure 3 and Table 6 summarize the effect of contextual noise at $\lambda = 0.75$ for $\sigma_Q \in \{0, 0.05, 0.10, 0.15, 0.20\}$. The three aggregation operators exhibit different sensitivity patterns. For the linear operator, noise increases an already existing vulnerability: V grows from 0.077 at $\sigma_Q = 0$ to 0.156 at $\sigma_Q = 0.20$. In contrast, the weighted geometric and minimum operators start from perfect compliance at $\sigma_Q = 0$, consistent with their zero-absorption behavior. However, this guarantee relies on the availability of accurate contextual information. When noise causes a true veto ($Q_i = 0$) to be observed as $Q'_i > 0$, the operator no longer identifies the alternative as forbidden. Such events are termed *false-negative contextual errors*. In the language of Corollary 2, noise breaks the hypothesis $Q_i = 0$ for the input actually supplied to the operator, so the exclusion guarantee no longer applies.

For the weighted geometric operator, every selected true veto in this design is a false negative; correspondingly, V increases from 0 to 0.046 at $\sigma_Q = 0.20$. The minimum operator is less affected, with $V = 0.002$ at the same noise level, because a positive contextual

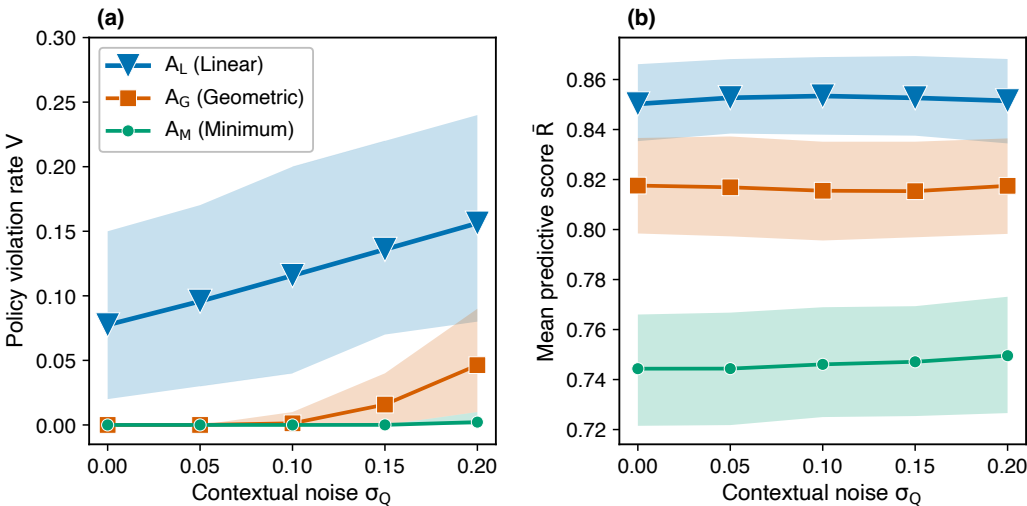


Figure 3. Sensitivity to contextual noise at $\lambda = 0.75$. Contextual scores are perturbed as $Q'_i = \text{clip}(Q_i + \epsilon_i, 0, 1)$, $\epsilon_i \sim \mathcal{N}(0, \sigma_Q^2)$; aggregation uses Q'_i while policy violations are evaluated using the true Q_i . Results are Monte Carlo means with 95% percentile intervals over 1000 replications ($N = 1000$, $K = 100$). (a) Policy violation rate $V(\sigma_Q)$. (b) Mean original predictive score $\bar{R}(\sigma_Q)$.

observation alone is insufficient to yield a high prioritization score: minimum aggregation requires both contributing dimensions to remain sufficiently large.

Table 6. Policy violation rate V (evaluated on true Q) and false-negative veto rate in Top- K under contextual uncertainty at $\lambda = 0.75$. Contextual observations are perturbed as $Q'_i = \text{clip}(Q_i + \epsilon_i, 0, 1)$ with $\epsilon_i \sim \mathcal{N}(0, \sigma_Q^2)$. Values are Monte Carlo means over 1000 replications ($N = 1000$, $K = 100$).

σ_Q	Operator	V	FN rate
0.00	A_L	0.077	0.000
	A_G	0.000	0.000
	A_M	0.000	0.000
0.05	A_L	0.096	0.063
	A_G	0.000	0.000
	A_M	0.000	0.000
0.10	A_L	0.116	0.090
	A_G	0.001	0.001
	A_M	0.000	0.000
0.15	A_L	0.136	0.117
	A_G	0.016	0.016
	A_M	0.000	0.000
0.20	A_L	0.156	0.145
	A_G	0.046	0.046
	A_M	0.002	0.002

The mechanism behind the loss of compliance is shown in Figure 4. The figure reports the fraction of selected alternatives that are true vetoes ($Q_i = 0$) but are observed with $Q'_i > 0$. For the weighted geometric operator, the false-negative rate matches the increase in V . Contextual uncertainty does not destroy the operator’s zero-absorption property with respect to Q'_i ; rather, it breaks preservation of vetoes defined on the unobserved true state Q_i . The minimum operator remains almost unaffected because false-negative contextual observations rarely produce scores large enough to enter the Top- K set.

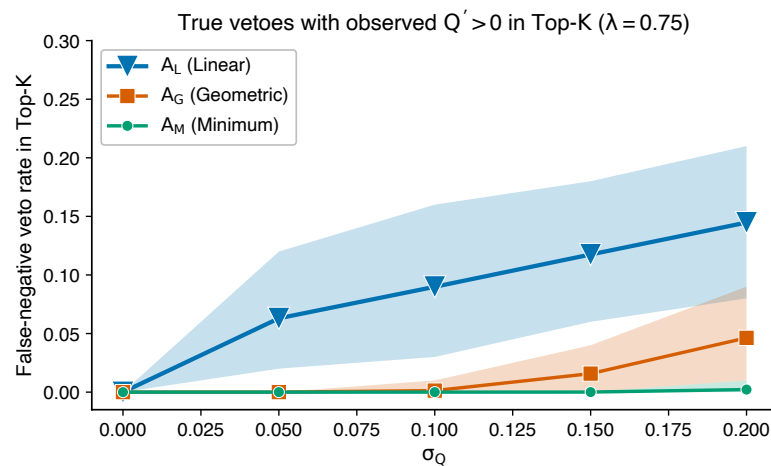


Figure 4. False-negative contextual errors at $\lambda = 0.75$: fraction of Top-K alternatives that are true vetoes ($Q_i = 0$) but are observed with $Q'_i > 0$ under contextual noise $Q'_i = \text{clip}(Q_i + \epsilon_i, 0, 1)$, $\epsilon_i \sim \mathcal{N}(0, \sigma_Q^2)$. Points are Monte Carlo means with 95% percentile intervals over 1000 replications ($N = 1000$, $K = 100$).

Finally, alternative synthetic populations indicate that the reported mechanisms are not confined to the baseline configuration. Changes in veto prevalence, predictive separation, and the distribution of low contextual scores alter the magnitude of violations. Detailed results, including the exact primary-setting values, are reported in Appendix B.1.

6.4. Sensitivity Analysis

Sections 6.1–6.3 established how the aggregation operators behave under contextual vetoes, predictive overconfidence, and imperfect contextual observation. Those experiments did not, however, quantify how stable the resulting prioritization decisions remain when the aggregation weight or the predictive scores are perturbed. The present experiment addresses this question directly. Stability is measured in two complementary ways, following the definitions in Section 5.4: Kendall's τ captures agreement of the complete ranking with a reference ranking, whereas the Top-K Jaccard index measures overlap of the selected decision set. The analysis uses the same baseline population and Monte Carlo protocol as the preceding experiments ($N = 1000$, $K = 100$, 1000 replications, seeds 42–1041), with reference rankings computed at $\lambda = 0.75$.

Figure 5 reports ranking and selection stability under local perturbations of the aggregation weight, $\lambda \in \{0.70, 0.725, 0.75, 0.775, 0.80\}$, with exact predictive scores. Rankings obtained at $\lambda = 0.75$ serve as the reference. The minimum operator is invariant over this grid ($\tau = 1$, Jaccard = 1). This is a direct consequence of the definition of A_M , which contains no weighting parameter, and is reported here as a consistency check on the implementation rather than as a result. For the same reason, the comparison in this first protocol is informative only for the two λ -dependent operators. The weighted geometric operator remains highly stable: the lowest observed values are $\tau = 0.965$ and Jaccard = 0.886 (at $\lambda = 0.70$ and $\lambda = 0.80$, respectively). The linear operator is more sensitive, with τ falling to 0.926 at $\lambda = 0.70$ and Jaccard to 0.743 at $\lambda = 0.80$. Within this neighbourhood the geometric operator is thus the less λ -sensitive of the two. Full numerical values are reported in Appendix C.

A second protocol introduces unbiased stochastic uncertainty in the predictive scores of all alternatives, $R'_i = \text{clip}(R_i + \epsilon_i, 0, 1)$ with $\epsilon_i \sim \mathcal{N}(0, \sigma_R^2)$ and $\sigma_R \in \{0, 0.02, 0.05, 0.10, 0.20\}$, holding $\lambda = 0.75$ fixed. Unlike the systematic overconfidence studied in Section 6.2, this perturbation has zero mean and is applied uniformly rather

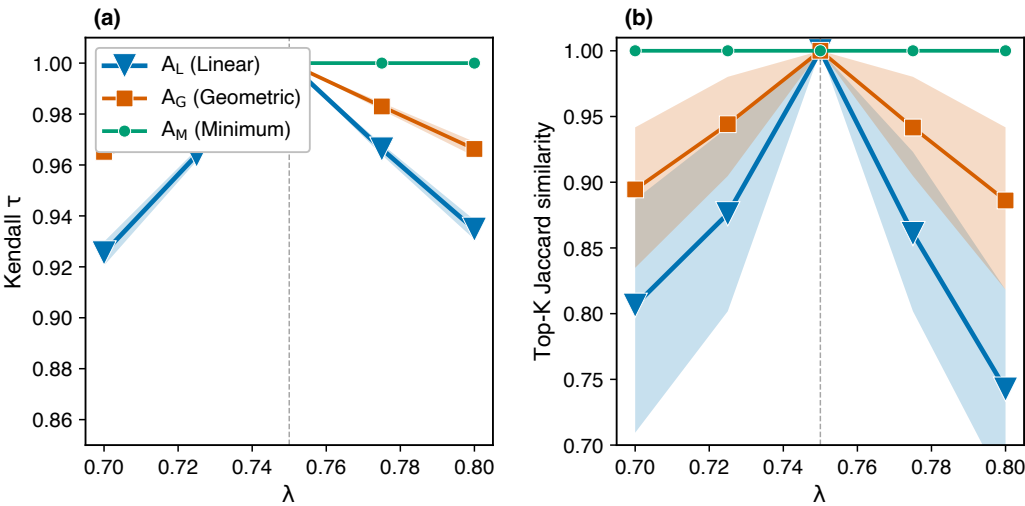


Figure 5. Ranking and Top-K stability under λ perturbation around the reference $\lambda_{\text{ref}} = 0.75$ ($\sigma_R = 0$, baseline population). Both metrics are computed relative to the ranking obtained at $\lambda = 0.75$. Results are Monte Carlo means with 95% percentile intervals over 1000 replications ($N = 1000$, $K = 100$). (a) Kendall’s $\tau(\lambda)$. (b) Top-K Jaccard similarity.

than only to contextually weak cases. The reference ranking is the noiseless ranking at $\sigma_R = 0$. Because the perturbation acts on R rather than on λ , this protocol places the three operators on an equal footing and therefore carries the comparative weight of the section. Figure 6 and Table 7 show that all three operators become less stable as σ_R increases, while the ordering $A_M > A_G > A_L$ is preserved throughout the grid. At $\sigma_R = 0.20$, Kendall’s τ is 0.545 for A_L , 0.705 for A_G , and 0.757 for A_M .

Table 7. Ranking and Top-K stability under unbiased predictive score uncertainty at $\lambda = 0.75$. Predictive scores are perturbed as $R'_i = \text{clip}(R_i + \epsilon_i, 0, 1)$ with $\epsilon_i \sim \mathcal{N}(0, \sigma_R^2)$, applied to all alternatives. Kendall’s τ and Top-K Jaccard similarity are computed relative to the noiseless ranking at $\sigma_R = 0$. Values are Monte Carlo means over 1000 replications ($N = 1000$, $K = 100$).

σ_R	Operator	Kendall τ	Jaccard Top-K
0.00	A_L	1.0000	1.0000
	A_G	1.0000	1.0000
	A_M	1.0000	1.0000
0.02	A_L	0.9442	0.8373
	A_G	0.9663	0.8948
	A_M	0.9727	0.9274
0.05	A_L	0.8629	0.6589
	A_G	0.9162	0.7620
	A_M	0.9321	0.8382
0.10	A_L	0.7390	0.4781
	A_G	0.8371	0.5960
	A_M	0.8677	0.7148
0.20	A_L	0.5454	0.3356
	A_G	0.7049	0.4021
	A_M	0.7572	0.5405

The two metrics do not degrade at the same rate. At $\sigma_R = 0.20$, Top-K Jaccard falls to 0.336 (A_L), 0.402 (A_G), and 0.541 (A_M), a larger relative reduction than that observed

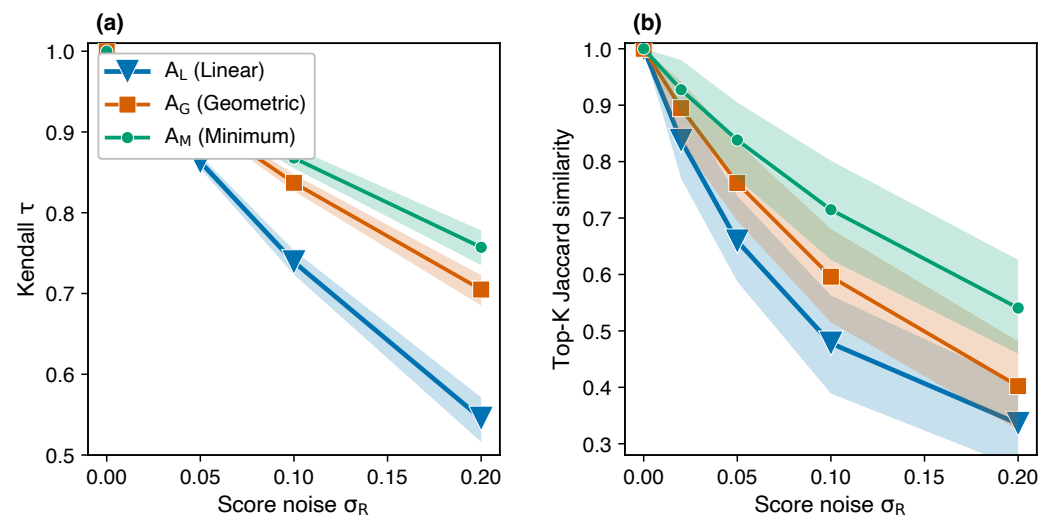


Figure 6. Ranking and Top-K stability under unbiased predictive score uncertainty at $\lambda = 0.75$. Predictive scores are perturbed as $R'_i = \text{clip}(R_i + \epsilon_i, 0, 1)$, $\epsilon_i \sim \mathcal{N}(0, \sigma_R^2)$, applied to all alternatives. Both metrics are computed relative to the noiseless ranking at $\sigma_R = 0$. Results are Monte Carlo means with 95% percentile intervals over 1000 replications ($N = 1000$, $K = 100$). (a) Kendall's $\tau(\sigma_R)$. (b) Top-K Jaccard similarity.

for Kendall's τ . This pattern indicates that predictive uncertainty has a particularly strong effect on membership of the selected Top-K set, rather than producing only local swaps deep in the ranking. The comparison is observational: it characterizes the sensitivity experiment and is not offered as a mechanistic proof of how individual alternatives enter or leave the decision set.

Within the tested perturbation ranges and the baseline synthetic population, A_G and especially A_M therefore yield more stable prioritization decisions than A_L . This is an empirical finding for the present setting, not a claim of universal operator superiority: it states that the contextual-aware aggregation rules were less sensitive to the tested parameter and predictive-score perturbations.

7. Employee Attrition Case Study

7.1. Original CADEMAs-ML Application

7.2. Experimental Setup

7.3. Results and Interpretation

8. Discussion

8.1. Implications for Cooperative Automated Decision Making

8.2. Relation to Fuzzy MCDM and Aggregation Theory

8.3. Limitations

9. Conclusions and Future Work

Appendix A. Effects of Predictive Overconfidence

This appendix supplements Section 6.2 with diagnostic analyses of the overconfidence perturbation protocol.

Appendix A.1. Clipped versus unclipped overconfidence

To assess whether the non-monotonic $V(\delta)$ curve for A_L at $\lambda = 0.75$ depends on bounding perturbed predictive scores, a diagnostic counterfactual was evaluated on identical populations and random seeds: an unclipped perturbation $R'_i = R_i + \delta$ for $Q_i \leq q_{\text{weak}}$, with all other protocol details unchanged. Table A1 reports the resulting policy violation

rates. Under clipping, $V(\delta)$ peaks at $\delta = 0.10$ and declines to 0.022 at $\delta = 0.30$. Without clipping, $V(\delta)$ increases monotonically over the same grid, reaching $V = 0.784$ at $\delta = 0.30$. Figure A1 compares $V(\delta)$ and total weak-context exposure E_{weak} under both models.

Table A1. Policy violation rate $V(\delta)$ for A_L at $\lambda = 0.75$ under the formal clipped model and an unclipped diagnostic counterfactual (same populations and seeds; $N = 1000$, $K = 100$, $q_{\text{weak}} = 0.25$, 1000 Monte Carlo replications).

δ	Clipped V	Unclipped V
0.00	0.077	0.077
0.05	0.209	0.210
0.10	0.251	0.351
0.15	0.196	0.488
0.20	0.131	0.611
0.25	0.068	0.712
0.30	0.022	0.784

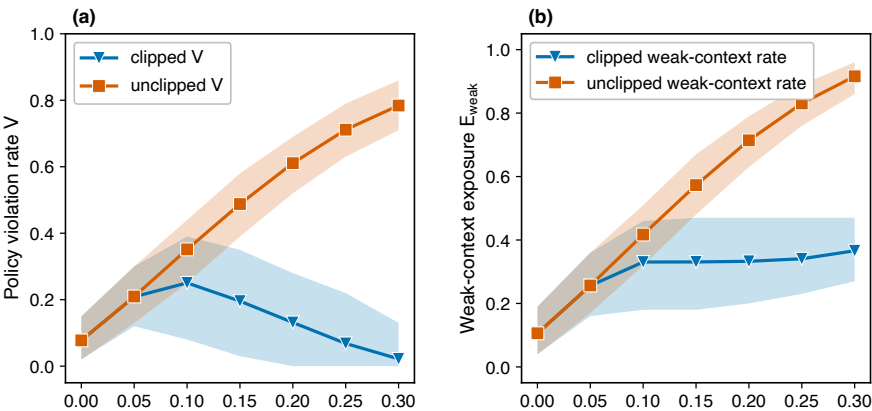


Figure A1. Diagnostic comparison of clipped and unclipped overconfidence at $\lambda = 0.75$. (a) Policy violation rate $V(\delta)$. (b) Weak-context exposure $E_{\text{weak}}(\delta) = |\{i \in \mathcal{T}_K : Q_i \leq q_{\text{weak}}\}|/K$. Same Monte Carlo protocol as Section 6.2.

Appendix A.2. Ranking stability under overconfidence

Figure A2 reports Kendall’s τ between the ranking at $\delta = 0$ and the ranking under each tested δ value at $\lambda = 0.75$. For A_L , τ decreases from 1.000 at $\delta = 0$ to 0.736 at $\delta = 0.30$, reflecting substantial reordering of the prioritization as overconfidence increases. The weighted geometric and minimum operators exhibit smaller decreases in τ over the same grid while maintaining $V = 0$.

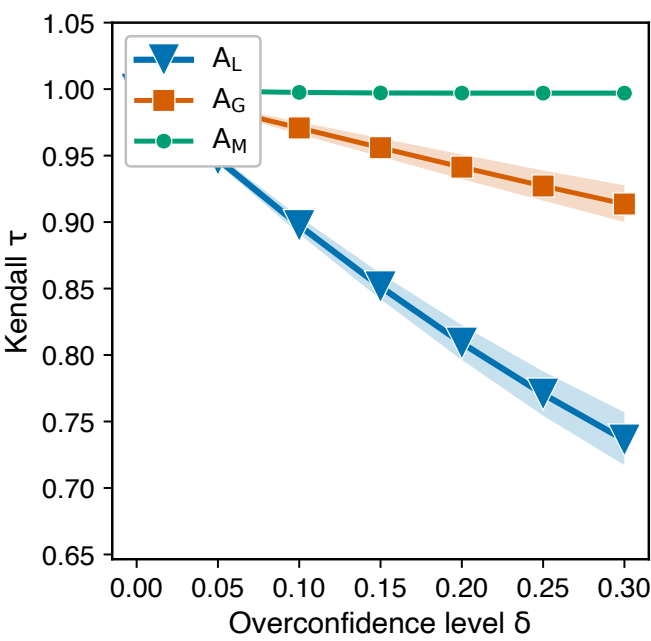


Figure A2. Kendall’s τ between the unperturbed ranking ($\delta = 0$) and the ranking under overconfidence δ , at $\lambda = 0.75$ ($N = 1000$, $K = 100$, $q_{\text{weak}} = 0.25$, 1000 Monte Carlo replications). Points are means; shaded bands are 95% percentile confidence intervals.

Appendix B. Contextual uncertainty diagnostics

This appendix provides additional robustness analyses supporting Section 6.3. The main experiment evaluates the effect of stochastic uncertainty in contextual assessments through the perturbation $Q'_i = \text{clip}(Q_i + \epsilon_i, 0, 1)$. This appendix examines whether the observed operator behaviour depends on the specific synthetic population used in the main experiments.

Appendix B.1. Population robustness variants

The robustness analysis considers alternative synthetic populations obtained by modifying three characteristics of the baseline generation process: the proportion of contextual vetoes, the prevalence of weak non-veto cases, and the overlap between predictive score distributions of vetoed and non-veto alternatives. The following variants were evaluated:

Table A2. Alternative population configurations used in the contextual uncertainty robustness analysis.

Scenario	Main modification with respect to baseline
Baseline	Original synthetic population configuration
Low veto fraction	Reduced proportion of contextual vetoes
High veto fraction	Increased proportion of contextual vetoes
Weak-heavy	Increased prevalence of weak non-veto alternatives
Weak-sparse	Reduced prevalence of weak non-veto alternatives
Low R separation	Reduced predictive-score separation between groups
High R overlap	Increased overlap between veto and non-veto predictive scores

For each population variant, the same Monte Carlo protocol described in Section 6.3 was applied ($N = 1000$, $K = 100$, and 1000 replications). The purpose of this analysis is not to identify a universally optimal operator, but to determine whether the qualita-

tive mechanisms observed in the baseline population persist under different population structures.

Table A3 summarizes the main results. The onset of policy violations for the linear operator remains in a similar range across populations, although the exact threshold depends on the relative frequency and characteristics of veto and weak-context alternatives. In particular, populations with a larger proportion of weak non-veto cases produce higher violation rates because more alternatives can receive non-zero contextual scores after aggregation.

Table A3. Robustness of the linear operator under alternative population configurations at $\sigma_Q = 0$. The table reports the first λ value with $V > 0$, together with the violation rate at $\lambda = 0.75$ and $\lambda = 1.0$.

Scenario	Onset λ	$V(0.75)$	$V(1.0)$
Baseline	0.70	0.107	0.644
High R overlap	0.80	0.007	0.124
High veto fraction	0.70	0.053	0.418
Low R separation	0.70	0.053	0.400
Low veto fraction	0.80	0.028	0.240
Weak-heavy	0.60	0.284	0.642
Weak-sparse	0.50	0.411	0.643

The same qualitative behavior is observed when contextual uncertainty is introduced. At $\lambda = 0.75$ and $\sigma_Q = 0.20$, the violation rate varies quantitatively across populations, ranging from $V = 0.011$ in the high-overlap scenario to $V = 0.456$ in the weak-sparse scenario. These differences reflect changes in the available alternatives rather than a change in the underlying aggregation mechanism.

Overall, the experiments confirm that the main conclusions of Section 6.3 are structural. The zero-absorption behavior of A_G and A_M under exact contextual information, and the emergence of violations for A_L when predictive evidence dominates or contextual observations become uncertain, persist across the tested population configurations.

Appendix C. Sensitivity-analysis diagnostics

This appendix supplements Section 6.4 with the complete λ -perturbation table and a compact mean/worst-case summary. The experimental protocol is identical to that of the main text: baseline synthetic population, $N = 1000$, $K = 100$, 1000 Monte Carlo replications, and reference rankings at $\lambda = 0.75$. The quantities reported here are Kendall's τ (complete ranking agreement) and Top- K Jaccard similarity (overlap of the selected decision set). Policy-compliance metrics are not reported.

Table A4 lists Kendall's τ and Jaccard similarity at each tested λ relative to the ranking at $\lambda = 0.75$. The minimum operator is identically stable ($\tau = 1$, Jaccard = 1) because A_M does not depend on λ . The weighted geometric operator remains above $\tau = 0.965$ and Jaccard = 0.886 over the whole grid. The linear operator is the most sensitive of the three, with the largest Jaccard drop at $\lambda = 0.80$ (0.743).

Table A4. Ranking and Top-K stability under λ perturbation around $\lambda_{\text{ref}} = 0.75$ ($\sigma_R = 0$, baseline population). Kendall's τ and Top-K Jaccard similarity are computed relative to the reference ranking at $\lambda = 0.75$. Values are Monte Carlo means over 1000 replications ($N = 1000$, $K = 100$).

λ	Operator	Kendall τ	Jaccard Top-K
0.700	A_L	0.9255	0.8064
	A_G	0.9652	0.8945
	A_M	1.0000	1.0000
0.725	A_L	0.9639	0.8760
	A_G	0.9827	0.9441
	A_M	1.0000	1.0000
0.750	A_L	1.0000	1.0000
	A_G	1.0000	1.0000
	A_M	1.0000	1.0000
0.775	A_L	0.9662	0.8609
	A_G	0.9830	0.9416
	A_M	1.0000	1.0000
0.800	A_L	0.9347	0.7425
	A_G	0.9663	0.8861
	A_M	1.0000	1.0000

Table A5 and Figure A3 summarize mean and worst-case Kendall's τ after excluding the reference configurations ($\lambda = 0.75$ and $\sigma_R = 0$), where agreement is 1 by construction. Under both perturbation types the observed ordering is $A_M > A_G > A_L$. Score uncertainty produces a larger degradation than local λ shifts: the worst-case τ at $\sigma_R = 0.20$ is 0.545 for A_L , 0.705 for A_G , and 0.757 for A_M .

Table A5. Mean and worst-case Kendall's τ under the two perturbation protocols of Section 6.4. Averages exclude the reference configuration ($\lambda = 0.75$ and $\sigma_R = 0$), where $\tau = 1$ by construction. Values are Monte Carlo means over 1000 replications ($N = 1000$, $K = 100$).

Operator	Mean τ (λ)	Worst τ (λ)	Mean τ (σ_R)	Worst τ (σ_R)
A_L	0.9476	0.9255	0.7729	0.5454
A_G	0.9743	0.9652	0.8561	0.7049
A_M	1.0000	1.0000	0.8824	0.7572

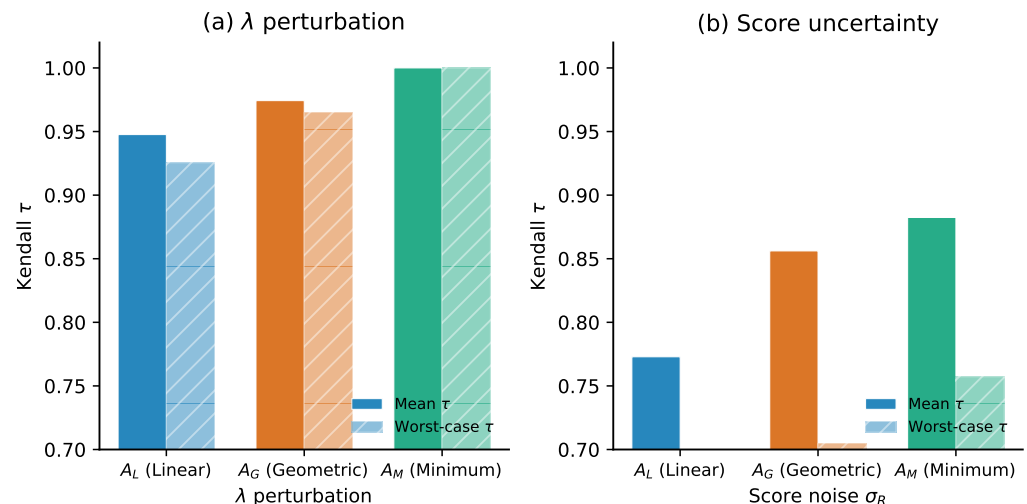


Figure A3. Mean and worst-case Kendall's τ by operator, excluding the reference configurations $\lambda = 0.75$ and $\sigma_R = 0$. (a) λ perturbation over $\{0.70, 0.725, 0.775, 0.80\}$. (b) Unbiased score uncertainty over $\sigma_R \in \{0.02, 0.05, 0.10, 0.20\}$. Same Monte Carlo protocol as Section 6.4.

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Conflicts of Interest: The authors declare no conflicts of interest.

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